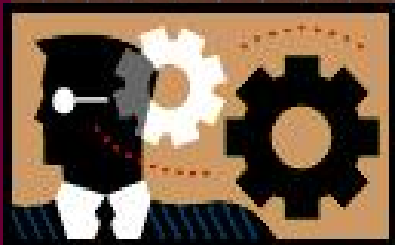




《线性代数》



第一章 线性方程组与行列式

§1.4 行列式的性质及应用

2025秋

复习回顾

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = \boxed{a_{11}a_{22}} - \boxed{a_{12}a_{21}}$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \boxed{a_{11}a_{22}a_{33}} + \boxed{a_{12}a_{23}a_{31}} + \boxed{a_{13}a_{21}a_{32}} \\ - \boxed{a_{13}a_{22}a_{31}} - \boxed{a_{12}a_{21}a_{33}} - \boxed{a_{11}a_{23}a_{32}}$$

复习回顾

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = \sum_{j_1 j_2 \cdots j_n} (-1)^{\tau(j_1 j_2 \cdots j_n)} a_{1j_1} a_{2j_2} \cdots a_{nj_n}$$
$$= \sum_{i_1 i_2 \cdots i_n} (-1)^{\tau(i_1 i_2 \cdots i_n)} a_{i_1 1} a_{i_2 2} \cdots a_{i_n n}$$
$$= \sum_{\substack{i_1 i_2 \cdots i_n \\ j_1 j_2 \cdots j_n}} (-1)^{\tau(i_1 i_2 \cdots i_n) + \tau(j_1 j_2 \cdots j_n)} a_{i_1 j_1} a_{i_2 j_2} \cdots a_{i_n j_n}$$

思考问题:

利用定义计算行列式是否简单?

什么情况下行列式的计算比较简单?





$$D = \begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix} = ?$$

$$\begin{aligned} D = & a_{11}a_{22}a_{33}a_{44} - a_{11}a_{22}a_{34}a_{43} - a_{11}a_{23}a_{32}a_{44} \\ & + a_{11}a_{23}a_{34}a_{42} + a_{11}a_{24}a_{32}a_{43} - a_{11}a_{24}a_{33}a_{42} \\ & - a_{12}a_{21}a_{33}a_{44} + a_{12}a_{21}a_{34}a_{43} + a_{12}a_{23}a_{31}a_{44} \\ & - a_{12}a_{23}a_{34}a_{41} - a_{12}a_{24}a_{31}a_{43} + a_{12}a_{24}a_{33}a_{41} \\ & + a_{13}a_{21}a_{32}a_{44} - a_{13}a_{21}a_{34}a_{42} - a_{13}a_{22}a_{31}a_{44} \\ & + a_{13}a_{22}a_{34}a_{41} + a_{13}a_{24}a_{31}a_{42} - a_{13}a_{24}a_{32}a_{41} \\ & - a_{14}a_{21}a_{32}a_{43} + a_{14}a_{21}a_{33}a_{42} + a_{14}a_{22}a_{31}a_{43} \\ & - a_{14}a_{22}a_{33}a_{41} - a_{14}a_{23}a_{31}a_{42} + a_{14}a_{23}a_{32}a_{41} \end{aligned}$$

例1

“洞” 

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{vmatrix} = a_{11}a_{22} \cdots a_{nn}.$$

上三角行列式

例2

$$\begin{vmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_{1n} \end{vmatrix} = d_1d_2 \cdots d_n.$$

对角行列式

例3

$$\begin{vmatrix} 0 & \cdots & 0 & a_{1n} \\ 0 & \cdots & a_{2(n-1)} & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & \cdots & a_{n(n-1)} & a_{nn} \end{vmatrix}$$

$$= (-1)^{\tau(n(n-1)\cdots 21)} a_{1n} a_{2(n-1)} \cdots a_{n1}$$

$$= (-1)^{\frac{n(n-1)}{2}} a_{1n} a_{2(n-1)} \cdots a_{n1}.$$

例4

$$\begin{vmatrix} 0 & \cdots & 0 & a_{1n} \\ 0 & \cdots & a_{2(n-1)} & 0 \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & \cdots & 0 & 0 \end{vmatrix}$$

$$= (-1)^{\frac{n(n-1)}{2}} a_{1n} a_{2(n-1)} \cdots a_{n1}.$$

例5

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ a_{31} & a_{32} & 0 & 0 & 0 \\ a_{41} & a_{42} & 0 & 0 & 0 \\ a_{51} & a_{52} & 0 & 0 & 0 \end{vmatrix} = 0.$$

思考问题:

利用定义计算行列式是否简单?

什么情况下行列式的计算比较简单?

答:

低阶矩阵、0元素多的矩阵、
上三角阵、下三角阵



内容提要

➤ 行列式的性质

➤ 行列式的计算

化繁为简

一. 行列式的性质

定义1 设

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}, \quad \text{记} \quad A^T = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & \cdots & a_{n2} \\ \vdots & \vdots & \vdots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{pmatrix},$$

称 A^T 为 A 的转置矩阵. $|A^T|$ 为 $|A|$ 的转置行列式.

一. 行列式的性质

性质1: 行列互换, 行列式的值不变, 即

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & \cdots & a_{n2} \\ \vdots & \vdots & \vdots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{vmatrix}.$$

转置不变, 行列等价

性质2:

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \vdots & \vdots \\ ka_{p1} & ka_{p2} & \cdots & ka_{pn} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = k \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{p1} & a_{p2} & \cdots & a_{pn} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}.$$

证明:

$$\begin{aligned} \text{左端} &= \sum_{j_1 j_2 \cdots j_n} (-1)^{\tau(j_1 j_2 \cdots j_n)} a_{1j_1} \cdots (ka_{pj_p}) \cdots a_{nj_n} \\ &= k \sum_{j_1 j_2 \cdots j_n} (-1)^{\tau(j_1 j_2 \cdots j_n)} a_{1j_1} \cdots a_{pj_p} \cdots a_{nj_n} = \text{右端}. \end{aligned}$$

倍乘可提

推论1. 若行列式的某行元素全为0, 则行列式等于0.

性质3:

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{p1} + a'_{p1} & a_{p2} + a'_{p2} & \cdots & a_{pn} + a'_{pn} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

行列加法，
拆项法则

$$= \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{p1} & a_{p2} & \cdots & a_{pn} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} + \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \vdots & \vdots \\ a'_{p1} & a'_{p2} & \cdots & a'_{pn} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}.$$

证明：

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{p1} + a'_{p1} & a_{p2} + a'_{p2} & \cdots & a_{pn} + a'_{pn} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$
$$= \sum_{j_1 j_2 \cdots j_n} (-1)^{\tau(j_1 j_2 \cdots j_n)} a_{1j_1} \cdots (a_{pj_p} + a'_{pj_p}) \cdots a_{nj_n}$$
$$= \sum_{j_1 j_2 \cdots j_n} (-1)^{\tau(j_1 j_2 \cdots j_n)} a_{1j_1} \cdots a_{pj_p} \cdots a_{nj_n}$$
$$+ \sum_{j_1 j_2 \cdots j_n} (-1)^{\tau(j_1 j_2 \cdots j_n)} a_{1j_1} \cdots a'_{pj_p} \cdots a_{nj_n} = \text{右端}$$

性质4: 对换行列式中两行的位置, 行列式反号, 即

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{p1} & a_{p2} & \cdots & a_{pn} \\ \vdots & \vdots & \vdots & \vdots \\ a_{q1} & a_{q2} & \cdots & a_{qn} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = - \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{q1} & a_{q2} & \cdots & a_{qn} \\ \vdots & \vdots & \vdots & \vdots \\ a_{p1} & a_{p2} & \cdots & a_{pn} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}.$$

对换取反

性质5: 若行列式中有两行成比例，则行列式为零，即

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{p1} & a_{p2} & \cdots & a_{pn} \\ \vdots & \vdots & \vdots & \vdots \\ ka_{p1} & ka_{p2} & \cdots & ka_{pn} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = 0.$$

同行化零，同比化零

证明: 当 $k=1$ 时, 由性质 4,

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{p1} & a_{p2} & \cdots & a_{pn} \\ \vdots & \vdots & \vdots & \vdots \\ a_{p1} & a_{p2} & \cdots & a_{pn} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = - \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{p1} & a_{p2} & \cdots & a_{pn} \\ \vdots & \vdots & \vdots & \vdots \\ a_{p1} & a_{p2} & \cdots & a_{pn} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = 0.$$

再由性质 2, 可得.

性质6: 把一行的某个倍数加到另一行，行列式的值不变，即

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{p1} + ka_{q1} & a_{p2} + ka_{q2} & \cdots & a_{pn} + ka_{qn} \\ \vdots & \vdots & \vdots & \vdots \\ a_{q1} & a_{q2} & \cdots & a_{qn} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{p1} & a_{p2} & \cdots & a_{pn} \\ \vdots & \vdots & \vdots & \vdots \\ a_{q1} & a_{q2} & \cdots & a_{qn} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}.$$

倍加不变

证明：

$$\text{左端} - \text{右端} = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \vdots & \vdots \\ ka_{q1} & ka_{q2} & \cdots & ka_{qn} \\ \vdots & \vdots & \vdots & \vdots \\ a_{q1} & a_{q2} & \cdots & a_{qn} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = 0.$$

二. 行列式的计算

例6

$$\begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \end{vmatrix}$$

“打洞法”

$$= \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -2 \end{vmatrix}$$

(各行减去第一行)

$$= -8.$$

例8 证明
$$\begin{vmatrix} a_1 + b_1 & b_1 + c_1 & c_1 + a_1 \\ a_2 + b_2 & b_2 + c_2 & c_2 + a_2 \\ a_3 + b_3 & b_3 + c_3 & c_3 + a_3 \end{vmatrix} = 2 \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}.$$

证明：

$$\text{左边} = \begin{vmatrix} 2(a_1 + b_1 + c_1) & b_1 + c_1 & c_1 + a_1 \\ 2(a_2 + b_2 + c_2) & b_2 + c_2 & c_2 + a_2 \\ 2(a_3 + b_3 + c_3) & b_3 + c_3 & c_3 + a_3 \end{vmatrix}$$

$$= 2 \begin{vmatrix} a_1 + b_1 + c_1 & b_1 + c_1 & c_1 + a_1 \\ a_2 + b_2 + c_2 & b_2 + c_2 & c_2 + a_2 \\ a_3 + b_3 + c_3 & b_3 + c_3 & c_3 + a_3 \end{vmatrix}$$

$$= 2 \begin{vmatrix} a_1 + b_1 + c_1 & b_1 + c_1 & c_1 + a_1 \\ a_2 + b_2 + c_2 & b_2 + c_2 & c_2 + a_2 \\ a_3 + b_3 + c_3 & b_3 + c_3 & c_3 + a_3 \end{vmatrix}$$

$$\underline{\underline{c_1 - c_2 \rightarrow c_1}} \quad 2 \begin{vmatrix} a_1 & b_1 + c_1 & c_1 + a_1 \\ a_2 & b_2 + c_2 & c_2 + a_2 \\ a_3 & b_3 + c_3 & c_3 + a_3 \end{vmatrix} \quad \underline{\underline{c_3 - c_1 \rightarrow c_3}} \quad 2 \begin{vmatrix} a_1 & b_1 + c_1 & c_1 \\ a_2 & b_2 + c_2 & c_2 \\ a_3 & b_3 + c_3 & c_3 \end{vmatrix}$$

$$\underline{\underline{c_2 - c_3 \rightarrow c_2}} \quad 2 \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}.$$

例9 计算 $D = \begin{vmatrix} b & a & \cdots & a \\ a & b & \cdots & a \\ \vdots & \vdots & \ddots & \vdots \\ a & a & \cdots & b \end{vmatrix}.$

证明：

$$D = \begin{vmatrix} b & a & \cdots & a \\ a & b & \cdots & a \\ \vdots & \vdots & \ddots & \vdots \\ a & a & \cdots & b \end{vmatrix} = \begin{vmatrix} b + (n-1)a & a & \cdots & a \\ b + (n-1)a & b & \cdots & a \\ \vdots & \vdots & \ddots & \vdots \\ b + (n-1)a & a & \cdots & b \end{vmatrix}$$

$$\begin{aligned}
 \text{证明 : } D &= \begin{vmatrix} b & a & \cdots & a \\ a & b & \cdots & a \\ \vdots & \vdots & \vdots & \vdots \\ a & a & \cdots & b \end{vmatrix} = \begin{vmatrix} b + (n-1)a & a & \cdots & a \\ b + (n-1)a & b & \cdots & a \\ \vdots & \vdots & \vdots & \vdots \\ b + (n-1)a & a & \cdots & b \end{vmatrix} \\
 &= [b + (n-1)a] \begin{vmatrix} 1 & a & \cdots & a \\ 1 & b & \cdots & a \\ \vdots & \vdots & \vdots & \vdots \\ 1 & a & \cdots & b \end{vmatrix} = [b + (n-1)a] \begin{vmatrix} 1 & a & \cdots & a \\ 0 & b-a & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & b-a \end{vmatrix} \\
 &= [b + (n-1)a](b-a)^{n-1}.
 \end{aligned}$$

本讲小结

➤ 行列式的性质

转置不变，行列等价
倍乘可提，对换取反
行列加法，拆项法则
最后记住，倍加不变

➤ 行列式的计算

思考问题：

以例9为例，能
否用数学归纳法
计算行列式？

$$D = \begin{vmatrix} b & a & \cdots & a \\ a & b & \cdots & a \\ \vdots & \vdots & \ddots & \vdots \\ a & a & \cdots & b \end{vmatrix} = [b + (n-1)a](b-a)^{n-1}$$

补充1

$$D = \begin{vmatrix} 1+x & 1 & 1 & 1 \\ 1 & 1-x & 1 & 1 \\ 1 & 1 & 1+y & 1 \\ 1 & 1 & 1 & 1-y \end{vmatrix}$$

补充2

$$D = \begin{vmatrix} 1 & 2 & 2 & \cdots & 2 \\ 2 & 2 & 2 & \cdots & 2 \\ 2 & 2 & 3 & \cdots & 2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 2 & 2 & 2 & \cdots & n \end{vmatrix}$$

补充3

$$\begin{vmatrix} a & & & & b \\ & \ddots & & & \\ & & a & b & \\ & & b & a & \\ & & & & \ddots \\ b & & & & & a \end{vmatrix} \quad (2n\text{阶})$$

本讲作业

- 教材33页5(2), 6(2)(3), 7(1)(2)(3)
- 参考书《线性代数入门》（纸质版）46页1.3.7, 172页4.2.6